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Original Article

Phenotypic and genotypic characterization of antibiotic-resistant *Salmonella* isolated from humans, aquaculture, and poultry in Sri Lanka: A retrospective study



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ABSTRACT

Background: In Sri Lanka, foodborne diseases caused by nontyphoidal *Salmonella* are of increasing concern. We therefore aimed to characterize the dominant *Salmonella* serovars in humans, poultry, and aquaculture through a One Health approach.

Methods: We collected isolates from different sectors, confirmed their identities using PCR, screened their antibiotic resistance profiles, and determined their antibiotic resistance genes based on whole-genome sequencing.

Results: Of the 75 *Salmonella* isolates identified, the majority of serotypes were unidentified. Both *Salmonella enterica* serovar Enteritidis (*S. Enteritidis*) and *Salmonella enterica* serovar Typhimurium (*S. Typhimurium*) could be isolated from human sources and were also found prevalent in the poultry sector. ST36, ST11 and ST1541 were the dominant serotypes of *S. Typhimurium* and *S. Enteritidis*, respectively. Alarming, 4% (1/25) of poultry *Salmonella* isolates were resistant to ciprofloxacin, suggesting an emergence of this phenotype. Moreover, virulence genes were very diverse among *S. Enteritidis* and *S. Typhimurium* isolates.

Conclusions: With the diversity of unidentified serotypes found and the detection of emerging resistances, our study highlights the importance of a One Health approach to monitoring antibiotic resistance. For public health initiatives in Sri Lanka to be successful in mitigating salmonellosis, all three sectors – humans, aquaculture, and poultry – must be tackled concomitantly in a coordinated manner under the One Health approach because antibiotic resistance genes, and even specific sequence types, may be able to spread across the aforementioned sectors. We anticipate that our results will inform public health policies in Sri Lanka to tackle foodborne illnesses.

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Introduction

The surveillance of antimicrobial-resistant bacteria is a global health priority as diseases become more difficult to treat with antibiotics [1]. Regular phenotypic and genotypic characterization of bacteria would help inform of trends in resistance and virulence, prevalence, and genetic relatedness [2].

Salmonella enterica has caused millions of foodborne illnesses and thousands of human deaths in both developing and developed countries [3]. With over 2500 *S. enterica* serovars, salmonellosis in humans is most commonly caused by *S. Typhimurium* and *S. Enteritidis* [4]. In Sri Lanka, salmonellosis can be frequently traced to commercial broiler farms, commercial layer farms, layer breeders, and broiler breeders [5,6]. Indeed, the non-motile *S. enterica* serovar Pullorum and *S. enterica* serovar Gallinarum were commonly isolated from chicks and adult chickens, respectively [5]. As such, Sri Lankan *Salmonella* control programs for poultry initially focused on non-motile *Salmonella* spp [6]. While these programs have been successful, the prevalence of salmonellosis caused by motile *Salmonella* spp. – *S. Enteritidis* and *S. Typhimurium* – has increased. *Salmonella* can be isolated from many different sources of food in Sri Lanka, including broiler chicken meat (11.6% prevalence) [7], shrimp (25.5%) [8], meat (17.7%) [9], and fish 5.6% [10]. However, transmission routes of *Salmonella* are poorly understood despite the threats of it and its antimicrobial resistances spreading to humans [11,12]. As such, the dominant serovars affecting humans, relative contributions of different foods as sources of *Salmonella*, and genotypic and phenotypic diversity of strains are not clearly understood in Sri Lanka. Indeed, *S. enterica* alone has more than 2500 serovars [13]. As such, we used whole-genome sequencing (WGS) to characterize the genotypes and phenotypes of *Salmonella* isolates across Sri Lanka using a One Health approach, aiming to find the prevalent serovar and resistances from each environment.

Materials and methods

Sample collection

With the help of the Sri Lanka Medical Research Institute, Veterinary Research Institute, and National Aquatic Research and Development Agency, 75 *Salmonella* isolates were collected between 2001 and 2002, with 25 isolates from each of the three sectors: humans, poultry, and aquaculture.

Human isolates were obtained from clinical fecal samples from hospitals located in Marawila, Kuliapitiya, Chilaw, Negambo, and Kalutara, which were archived at the reference laboratory of Medical Research Institute in Sri Lanka. Aquaculture (shrimp) samples were collected from shrimp farms in the Chilaw district. Poultry (chicken meat) samples were collected from live bird markets in Kuliapitiya, Dambadeniya, Pannala, Kurunegala, Mawanella, Kegalle, Matale, Akurana, Katugasthota, and Gampola (Fig. S1).

Bacterial culturing, isolation, and multiplex PCR

Salmonella was identified and isolated by following the ISO 6579:2002 standard [14]. Briefly, the samples were first pre-enriched in buffered peptone water (Oxoid, United Kingdom) and incubated at 37 °C for 18 h. Subsequently, samples were enriched in Rappaport Vassiliadis broth (Oxoid, United Kingdom) and incubated at 41.5 °C for 24 h. After incubation, a loop full of material from the selective media was streaked onto xylose lysine deoxycholate agar (Oxoid, United Kingdom). and incubated at 37 °C for 24 h. Finally, presumptive *Salmonella* positive colonies were cultured on nutrient agar, and biochemical tests were then performed to identify *Salmonella* isolates.

All isolates were stored at – 80 °C in Tryptose Soy Broth with 20% glycerol until further testing. Bacterial DNA was extracted using QIAamp® DNA Mini Kit (catalog no. 51304, QIAGEN GmbH, Germany) according to manufacturer's protocol. Extracted DNA was stored at – 80 °C until further testing.

Multiplex PCR was performed in a C1000 Touch Thermal Cycler (BIO-RAD, USA) to identify specific serovars [15,16] using the QIAGEN Multiplex PCR Kit (Qiagen, Germany) and primers detailed in Table

S1. PCR was carried out using HotStarTaq DNA Polymerase (Qiagen, Germany) according to the manufacturer's instructions first using primers specific to *Salmonella* spp. and *S. Enteritidis*. Samples negative for *S. Enteritidis* were then subject to PCR using primers specific to *S. Typhimurium*. Gel electrophoresis for amplicons were conducted on 2% agarose gel with a 100 bp DNA ladder.

Antimicrobial susceptibility testing

Antibiotics selected for testing were based on the records of the Veterinary Drugs Control Authority of the Department of Animal Production and Health (antibiotics used for aquaculture and poultry) and the records of the Medical Supplies Division of the Department of Health (antibiotics for human use). Per the WHO AWaRe classification, the selected antibiotics were grouped as 'Access group' (Doxycycline 30 µg (DO), Trimethoprim 1.25 µg/sulfamethoxazole 23.75 µg (SXT)) and 'Watch group' (Cefotaxime 30 µg (CTX), Ciprofloxacin 5 µg (CIP), Ceftriaxone 30 µg (CRO), Enrofloxacin 5 µg (ENR), Erythromycin 15 µg (ERY), Imipenem 10 µg (IPM), Meropenem 10 µg (MEM)) [17]. Antimicrobial susceptibility was determined using disk diffusion tests according to the Clinical and Laboratory Standards Institute [18,19].

Whole-genome sequencing of *Salmonella* isolates

WGS was conducted by MacroGen, Korea (Illumina Nextera DNA XT). Adapters were removed by BBTools [20], and reads were with a minimum length < 35 bp filtered out. FastQC [21] was used for reads quality assessment. Whole-genome assembly was carried out using Bactopia [22], a flexible pipeline for bacterial genome analysis.

Genome annotation and phylogeny analysis

Genes conferring resistance to antibiotics, virulence genes, and plasmid replicons were identified using ABRicate [23]. MlST (<https://github.com/tseemann/mlst>) was used to identify sequence types using the *S. enterica* multilocus sequence typing scheme (MLST). SISTR [24] was used to predict *Salmonella* serovars.

Genome assemblies were annotated using Prokka [25]. The core and accessory genomes were identified by Roary [26] using the GFF3 files generated by Prokka [25]. Genes that were found in all *Salmonella* isolates were classified as the core genome. FastTree [27] was used to generate maximum likelihood phylogenetic tree from the core genome alignment. The phylogenetic clusters were identified by TreeCluster [28] with default parameters. The tree was visualized using ggtree [29].

Results

Salmonella prevalence and the serovar distribution

A total of 75 *Salmonella* isolates were obtained from aquaculture, human, and poultry sources in Sri Lanka, of which 25 isolates were collected from each source. Among the 75 *Salmonella* isolates, 18 serovars were detected using SISTR, and two isolates were not serotyped (Table S2). Additionally, six *S. Enteritidis* isolates were classified into different serovars, including Corvallis (n = 4), Weltevreden (n = 1), and one unidentified serovar. Furthermore, three *S. Typhimurium* isolates were identified as *S. enterica* serovar Kentucky (n = 1), *S. enterica* serovar Braenderup (n = 1), and an unidentified serovar. These discrepancies indicated a degree of error in genosertotyping using WGS data. To more focused on the analysis of *S. Enteritidis* and *S. Typhimurium* isolates, serotypes that could not be assigned to *S. Enteritidis* or *S. Typhimurium* were classified as 'Other' (Fig. 1A). In all three sources, the 'Other' serotype was the most common. The prevalence of *S. Enteritidis* was higher in poultry and

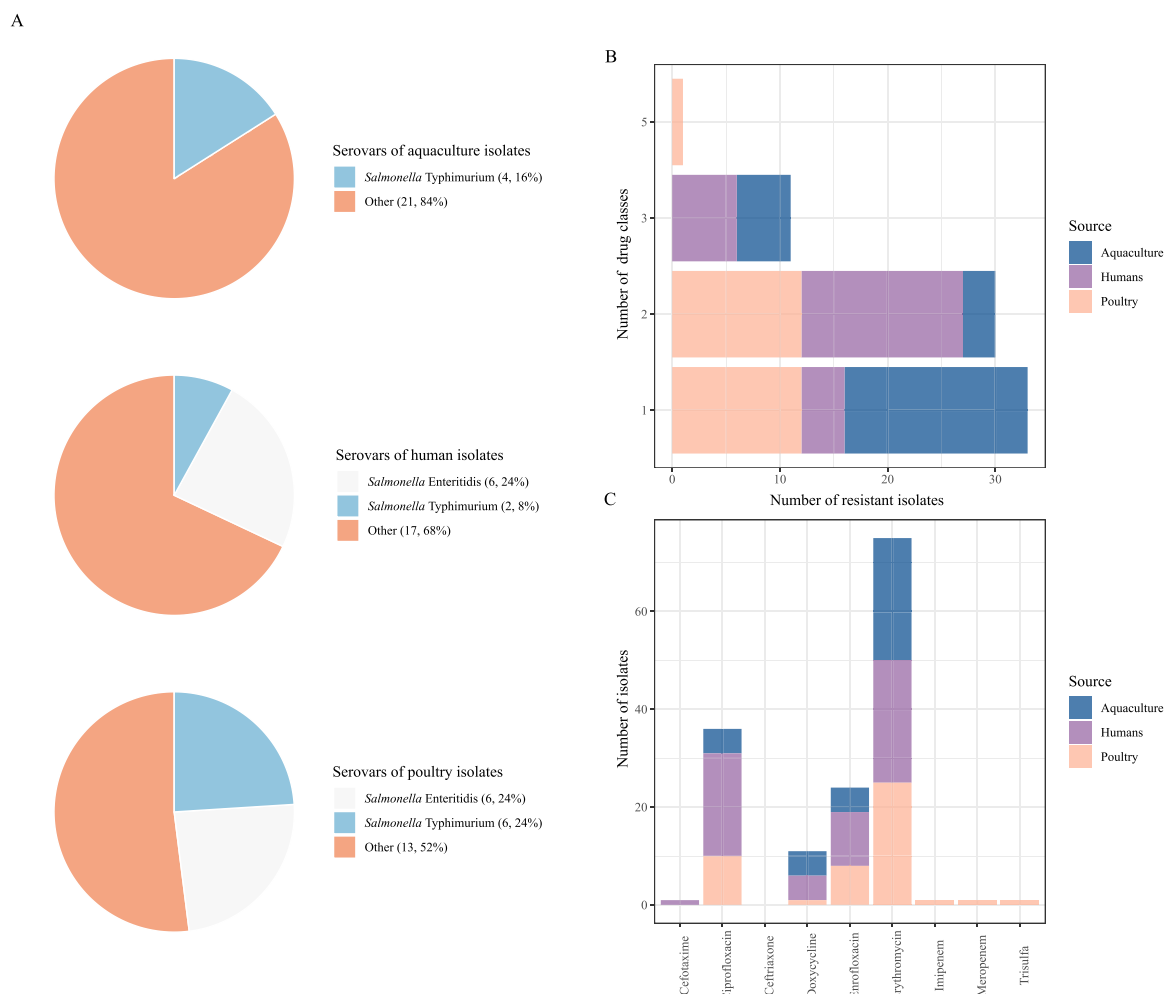


Fig. 1. The prevalence of different *Salmonella* serotypes and antibiotic resistance phenotypes among different sources. (A) The distribution of *Salmonella* serotypes. (B) Numbers of *Salmonella* isolates resistant to different antibiotic classes tested. (C) Numbers of isolates resistant to different antibiotics.

Table 1

The comparison of the proportion of serovars between different sources, *P* value was given by Fisher's exact test.

Serovars	Human (N = 25)	Poultry (N = 25)	Aquaculture (N = 25)	P value		
				Human vs aquaculture	Human vs poultry	Aquaculture vs poultry
<i>Salmonella</i> Enteritidis	6 (24%)	6 (24%)	0 (0%)	0.02229	1	0.02229
<i>Salmonella</i> Typhimurium	2 (8%)	6 (24%)	4 (16%)	0.6671	0.2467	0.7252

human samples than that in aquaculture ($P < 0.05$, Fisher's exact test) (Table 1).

Antibiotic resistance phenotype and genotype of *Salmonella* isolates

Of the nine antibiotics tested, all 75 *Salmonella* isolates were resistant to erythromycin but susceptible to ceftriaxone (Fig. 2). Twelve isolates (16.0%) were classified as multi-drug resistant (MDR, resistant to $\geq$ three drug classes) (Fig. 1B). Resistance to ciprofloxacin and/or enrofloxacin was seen in isolates from all sources. Regarding isolates from specific sources, all poultry isolates were resistant to fewer than two antimicrobials, except for one isolate resistant to five antimicrobial classes. Moreover, only poultry isolates were resistant to carbapenem (either meropenem or imipenem). Overall, however, human isolates accounted for a greater proportion of the resistances detected compared to isolates from other sources (Fig. 1B).

The complete genomes of all isolates were screened for known antimicrobial resistance (AMR) determinants (Fig. 4). AMR-conferring genes were detected in all isolates. Fifteen isolates – six aquaculture, seven human, and two poultry – were predicted to be multi-AMR (resistant to ≥ 3 drug classes). Almost all isolates ($n = 72$, 96.0%) were predicted to harbor the *aac(6')-Iaa* gene that confers resistance to aminoglycosides. The *qnrS1* genes conferring resistance to quinolones were found in 15 (20.0%) isolates, while 12 (16.0%) isolates harbored the *tet(A)* gene, which confers resistance to tetracyclines. Genes encoding β -lactamases (*bla*CMY-53, *bla*DHA-1, *bla*OXA-1, *bla*TEM-1B) were rarely detected in our isolates.

MLST, plasmid, and virulence gene profiles of *Salmonella* isolates

Based on MLST, the most prevalent sequence type was ST1541 ($n = 12$, 16.0%), which mostly consisted of *S. Enteritidis* isolates ($n = 4$, 33.3%) and isolates of unknown serovars ($n = 8$, 66.7%) from aquaculture, humans, and poultry (Fig. 3).

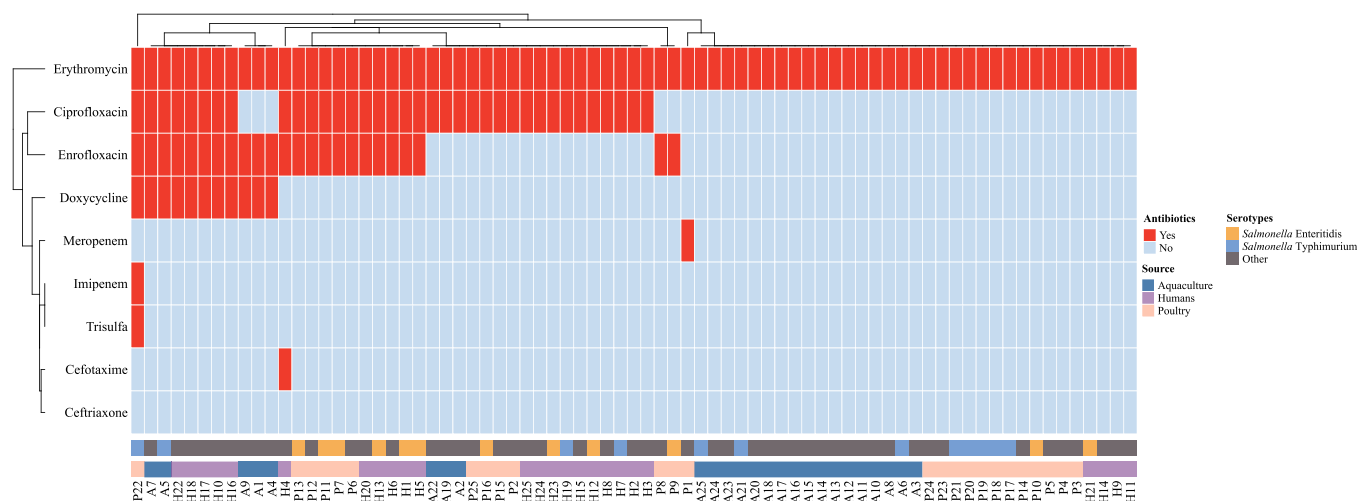


Fig. 2. Phenotypic antibiotic resistance profiles of all 75 *Salmonella* isolates from different sources.

ST11 ($n=6$, 50.0%) and ST1541 ($n=4$, 33.3%) were the most common sequence types in *S. Enteritidis* isolates ($n=12$, 16.0%). ST36 ($n=9$, 75.0%) was predominant in *S. Typhimurium* isolates ($n=12$, 16.0%). Nearly half of the MDR isolates ($n=12$, 16.0%) were of sequence type ST1541 ($n=5$, 41.7%). The remaining isolates were of sequence types ST198 ($n=4$, 33.3%), ST43 ($n=1$, 8.3%), ST314 ($n=1$, 8.3%), and one unknown sequence type ($n=1$, 8.3%).

The most common plasmids present in the isolates were ColRNAI_1 ($n=20$, 26.7%), IncFII(S)_1 ($n=19$, 25.3%), and Col440I_1 ($n=18$, 24%) (Fig. 3). Interestingly, the IncFIB(S)_1 plasmid was only detected in human isolates, whereas Col440I_1, Col440I_1, ColRNAI_1, and IncFII(S)_1 plasmids were found in isolates from all three sources. Moreover, isolates of sequence type ST1541 tended to have more plasmids than isolates of other sequence types, with an average of 3 plasmid replicons per isolate.

To characterize the virulence of our isolates, we detected a total of 128 virulence genes across all our isolates using ABRicate [23] (Fig. 3). We divided these genes into eight categories based on function: effector delivery system, adherence, nutritional/metabolic factor, motility, anti-virulence, antimicrobial activity/competitive

advantage, immune modulation, and stress survival. The typhoid-associated virulence gene *cdtB* was found in eight isolates, none of which were *S. Enteritidis* nor *S. Typhimurium*. Meanwhile, the Fimbriae-encoding genes *pef* and the serum resistance genes *rck* were only detected in *S. Enteritidis* isolates. The *sodCI* gene and *spv* genes (*spvB*, *spvC*, and *spvR*), which contribute to the virulence of non-typhoid *Salmonella* strains, were detected in 38 isolates and five *S. Enteritidis* isolates, respectively. The anti-virulence gene *grvA* can be found in 29 isolates.

Pan-genome and phylogenetic analyses

Phylogenetic analyses were performed to determine the relatedness of *Salmonella* isolates from different sources (Fig. 4). A total of 18363 genes were identified based on pan-genome analysis. Of these, core genes (those present in $\geq 95\%$ of isolates) made up 19.0% ($n=3483$) of the entire pan-genome. The shell genes ($n=1821$, 15% $\leq$ presence in isolates $< 95\%$) and cloud genes ($n=13059$, present in $< 15\%$ isolates), which make up the accessory genome, constituted 81% of the pan-genome.

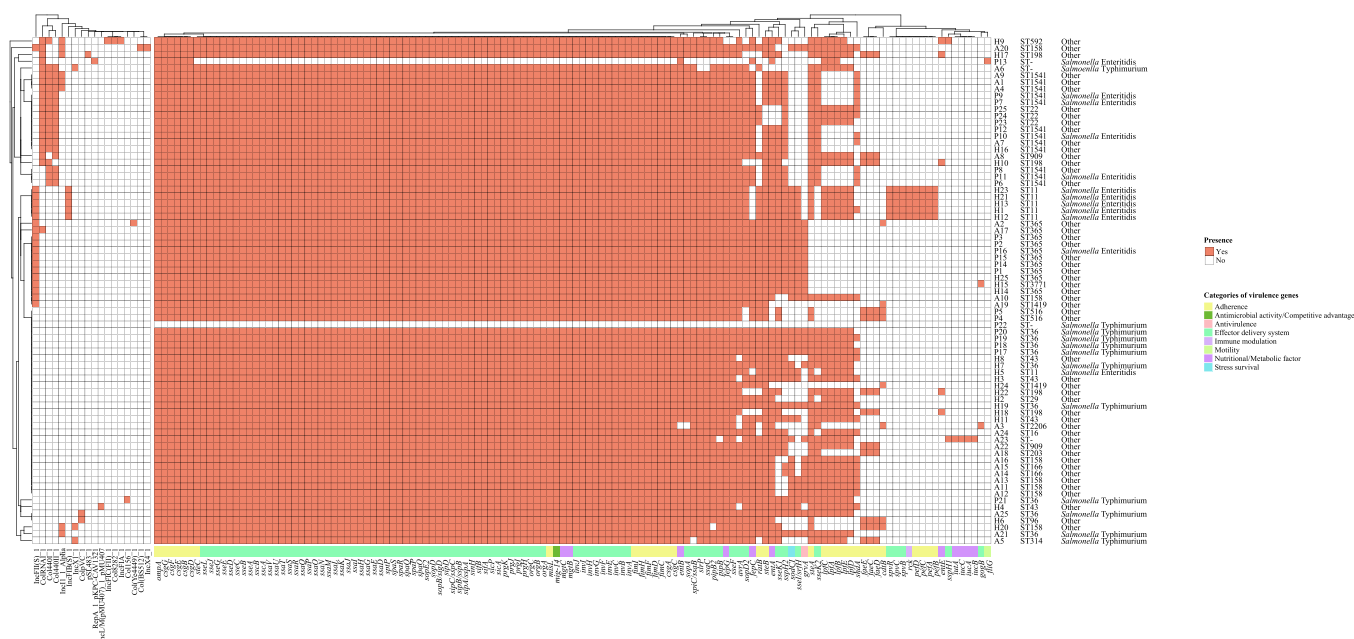


Fig. 3. Heatmap of the sequence types, plasmids, and virulence genes of the *Salmonella* isolates.

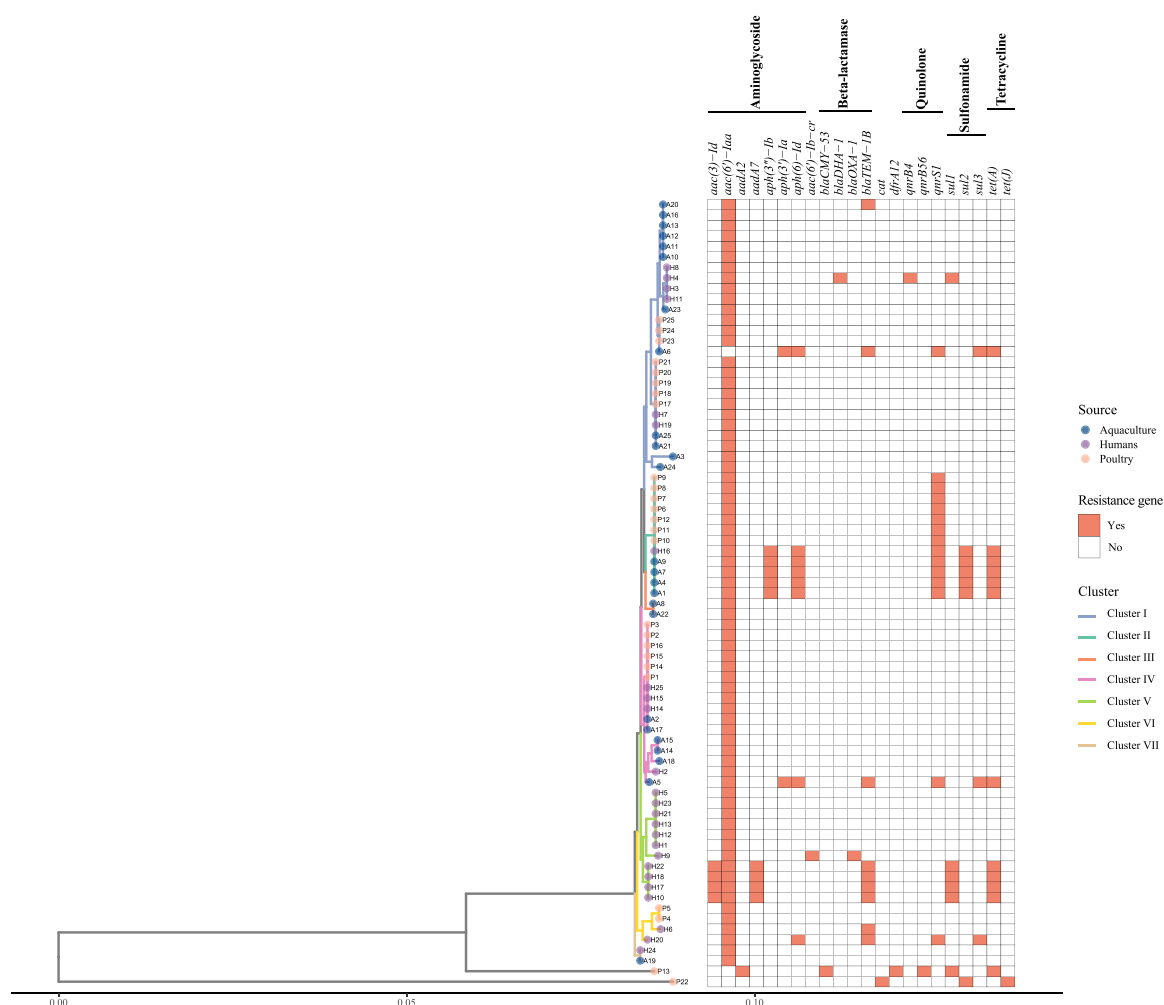


Fig. 4. Midpoint-rooted maximum likelihood phylogenetic tree based on the core gene alignment of all *Salmonella* isolates.

Phylogenetic analysis of all *Salmonella* isolates based on core genome alignment indicated that isolates from different sources were closely related, suggesting that a possible transmission event of antibiotic resistance genes may have occurred among these isolates (Fig. 4). The tree can be mainly divided into 7 distinct clusters by TreeCluster [28]. In cluster IV, isolates from distinct sources share the same pattern of AMR genes. Cluster V only contained human isolates, while other clusters contained isolates from a range of sources.

Discussion

Herein we characterize the different *Salmonella* serotypes we isolated from human, aquaculture, and poultry sources in Sri Lanka, including the genotypic and phenotypic AMR profiles of the isolates. Over 68% of the serovars in human, poultry, and aquaculture isolates were unidentifiable; *S. Enteritidis* and *S. Typhimurium* were not the dominant serotypes of isolates from any source, suggesting that, among food sources, other serovars may have a greater impact on public health. It is therefore concerning that current *Salmonella* control initiatives in Sri Lanka overlook these 'other' serovars. Regular, comprehensive surveillance is required to monitor the emergence of novel serovars across the three sources and better characterize them. In our study, *S. Enteritidis* isolates, which is commonly detected in human outbreaks, were primarily linked to human and poultry sources. Conversely, *S. Typhimurium* was identified from all three sources, indicating that risk of foodborne

Salmonella infections may arise from both poultry and aquaculture. Host-specific *Salmonella*, such as *S. Pullorum* and *S. Gallinarum*, were not isolated in this study. Given that contaminated environments in which poultry or fish are reared are considered major sources of foodborne *Salmonella* [30,31], improving the biosecurity and hygienic conditions of the sources we identified would likely greatly reduce the incidence of salmonellosis in Sri Lanka.

Salmonella infections are typically treated with third generation cephalosporins, quinolones, and macrolides [12,32]. Here we found that all *Salmonella* isolates that originated from poultry and aquaculture were entirely susceptible to WHO 'Watch' group antibiotics cefotaxime and ceftriaxone (third generation cephalosporins). Alarmingly, there appeared to be the emergence of resistance to the fluoroquinolone ciprofloxacin, to the quinolone enrofloxacin, to tetracyclines, and to carbapenems in poultry. While quinolones and tetracyclines are commonly used in the poultry industry in Sri Lanka [33,34], carbapenem is not. The emergence of carbapenem resistance was also seen in samples from aquaculture, despite these antibiotics not being used in this sector [34]. Therefore, the exact causes of carbapenem resistance here need to be further investigated to determine potential pathways of dissemination. Worryingly, in human isolates, we also detected emerging resistance to cefotaxime, an antibiotic used to treat severe multidrug resistant *Salmonella* infections [35].

Genotypic analysis showed that the most frequently harbored antibiotic resistance gene was *aac(6)-Iaa*, followed by *qnrS1* and *tet* (A), which confers resistance to aminoglycosides, quinolones, and

tetracyclines, respectively. However, *aac(6′)-Iaa* is rarely expressed and only confers aminoglycoside resistance in *Salmonella* in a handful of isolates [36]. We also found that 13 samples isolated across all three sources were multi-AMR, carrying AMR genes conferring resistance to more than two drug classes, indicating the potential risk of multi-drug resistant strains emerging in these environments. The majority of isolates shared a similar core genome, and the size of the accessory genome, which may contain high numbers of antibiotic resistance genes, varied in around 20% of isolates. Through phylogeny analysis, we divided our isolates into seven clusters. In cluster IV, isolates from three different sources were closely related, indicating that potential transmission events of antibiotic resistance genes may have occurred. In addition, genes conferring resistance to sulfamethoxazole were found in two unique clusters. Only two poultry isolates did not cluster together with the rest.

Among our isolates, the most common sequence type of *S. Typhimurium* was ST36. For *S. Enteritidis*, ST11 and ST1541 were the dominant sequence types. ST11 is the predominant sequence type in several countries [37–39] and has been isolated from many sources such as poultry, humans, and food, indicating that ST11 may have spread worldwide from a common clonal origin. We also detected ST1541 in all sources and isolates of this sequence type had the largest number of plasmid replicons. ST1541 and ST198 were also the most prevalent sequence type for MDR isolates.

Plasmid predictions indicated that there were 18 different plasmid replicons among our samples, with dominant replicons being ColRNAI_1, followed by IncFII(S)_1 and Col440I_1. Col440II_1, Col440I_1, ColRNAI_1, and IncFII(S)_1 plasmids were found in isolates from different sources, indicating that these mobile elements are able to spread across different sectors. Worryingly, IncFII(S)_1, which belongs to the IncF plasmid family, is associated with the dissemination of antimicrobial resistance and virulence [40]. Virulome analysis revealed eight isolates from all three different sources, none of which were *S. Enteritidis* nor *S. Typhimurium*, carried *cdtB*, which contributes to toxin production in non-typhoidal *Salmonella*. Other virulence factors were also detected, such as *pef* and *rck*, both of which were only detected in *S. Enteritidis* isolates. We also detected the *sodCI* gene and *spv* genes (*spvB*, *spvC*, and *spvR*).

However, there are several limitations in our study. Given its retrospective nature, this study is constrained by sample size and the focus on long-term, regular, and sector-specific sampling, impeding a comprehensive grasp of dynamic antimicrobial resistance changes across diverse sectors and the direct applicability of our findings to Sri Lanka's current landscape. Future investigations could leverage our results alongside more recent data to strengthen localized antibiotic resistance control strategies. Another limitation arises from the extensive diversity of *Salmonella* serovars, exceeding 2600 in number. Traditional methods like the Kauffman-White serotyping scheme [41] are time-consuming and costly, and using the WGS method for serovar identification carries a certain error rate. Thus, we utilized multiplex PCR, focusing solely on common serovars *S. Typhimurium* and *S. Enteritidis*, which are known to cause human salmonellosis. Nonetheless, other unidentified serovars have also exhibited antimicrobial resistance phenotypes in our study, indicating the potential hazards associated with *Salmonella* serovars beyond those examined.

Therefore, establishing a comprehensive regulatory framework is crucial. While Sri Lanka has taken measures and initiatives to control AMR in recent years, such as The Antibiotic Resistance Surveillance Project [42] launched in 2009, primarily focused on hospital samples, and the national guidelines [43] introduced in 2016 regarding the empirical and prophylactic use of antimicrobials, there is still a lack of oversight across human, animal, and environmental sectors. Conducting in-depth research on the antimicrobial resistance profiles of unidentified serovars through the combination of WGS and

PCR techniques and establishing empowering rules and regulations will significantly improve our ability to monitor and control the spread of antibiotic resistance.

Conclusion

Our study investigated antimicrobial resistance phenotypes and genotypes among *Salmonella* strains from humans, aquaculture, and poultry. *S. Enteritidis* was more prevalent in humans and poultry compared to aquaculture in our data. Additionally, all strains exhibited resistance to erythromycin, with 16% of the isolates displaying multidrug resistance. Isolates originating from humans were resistant to more antibiotics compared to isolates from the other two sectors. Through phylogenetic analysis, we observed that isolates from the three sectors were phylogenetically close to each other, suggesting possible transmission between different sources. Future studies are encouraged to embrace the One Health method for a comprehensive profiling of AMR and the development of intervention strategies across various sectors.

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CRediT authorship contribution statement

Conceptualization: K.H.U. and H.M.T.; methodology: K.H.U., R.P.M., and H.M.T.; software: K.H.U., R.P.M., and L.X.; validation: H.M.T.; formal analysis: K.H.U., R.P.M., and L.X.; investigation: K.H.U., S.P., and S.A.; resources: K.H.U. and H.M.T.; data curation: K.H.U., S.P., and S.A.; manuscript writing: K.H.U., R.P.M., and L.X.; reviewing and manuscript editing: L.X., M.K.L.W., and H.M.T.; data visualization: L.X.; project administration: H.M.T.; funding acquisition: H.M.T. All authors have read and agreed to the finalized version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jiph.2023.10.028](https://doi.org/10.1016/j.jiph.2023.10.028).

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